we can use this function three times to test whether a triangle is right-angled. square() is a "black box" — we need not know how the function computes the square. We only need to know its specification.

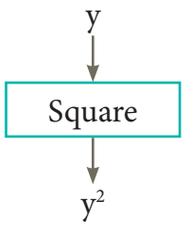


Figure 7.11: square function

```
1   right_angled(a, b, c)
2   -- inputs: c ≥ a, c ≥ b
3   -- outputs: result = true if c² = a² + b²;
4   --                result = false , otherwise
5   if square (c) = square (a) + square (b)
6      result := true
7   else
8      result := false
```

Points to Remember

- Compound statements are composed of sequential, alternative and iterative control flow statements.
- The value of a condition is true or false, depending on the values of the variables.
- Alternative statement selects and executes exactly one of the two statements,depending on the value of the condition.
- Conditional statement is executed only if the condition is true. Otherwise, nothing is done.
- Iterative statement repeatedly evaluates a condition and executes a statement as long as the condition is true.
- Programming language, pseudo code, and flowchart are notations for expressing algorithms.
- Decomposition breaks down a problem into smaller subproblems and combine their solutions to solve the original problem.
- A function is an abstraction of a subproblem, and specified by its input property, and its input-output relation.
- Users of function need to know only what the function does, and not how it is done.
- In refinement, starting from high level, each statement is repeatedly expanded into more detailed statements in the subsequent levels.

Evaluation

SECTION – A

Choose the correct answer

1. Suppose u, v = 10 ,5 before the assignment. What are the values of u and v after the sequence of assignments?

   1 u := v

   2 v := u

Chapter 7 Page 088-101.indd 98 3/24/2020 9:14:50 AM